Abhijeet Rathore

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Education

- **Indian Institute of Technology Roorkee** — Roorkee, India
  PhD - Dept. of Hydro and Renewable Energy; CGPA: 9.415 — *July 2024 - Present*
- **Energy Institute, Bengaluru (center of RGIPT)** — Bengaluru, India
  Master of Technology - Power and Energy Systems Engineering; CPI: 9.15 — *August 2020 - June 2022*
- **SRMCEM (Affiliated to Dr. A.P.J. Abdul Kalam Technical University)** — Lucknow, India
  Bachelor of Technology - Electrical and Electronics Engineering; Percentage: 77.48 — *July 2013 - June 2017*
- **10+2 (Intermediate)** — Lucknow, India
  CBSE Board; Percentage: 80% — *July 2012 - June 2013*
- **High School** — Lucknow, India
  CBSE Board; CGPA: 8.2 — *July 2010 - June 2011*

Experience

- **Research Engineer** — IIT Roorkee
  July 2022 – July 2024 — *Roorkee, India*
  - Developed a novel day-ahead photovoltaic (PV) power forecasting model, improving prediction accuracy by 67.9% with a 95% accuracy, enabling more reliable solar energy integration.
  - Managed end-to-end data workflows from six geographically distributed sites, including data acquisition, cleaning, processing, and model training to ensure high-quality input for machine learning pipelines.
  - Automated data collection and preprocessing pipelines, reducing manual intervention by over 90% and improving data accuracy by 95%.
  - Engineered interactive and user-friendly web interfaces using Flask and Streamlit frameworks, enhancing user engagement and accessibility of forecasting tools.
  - Led backend development for a commercial website, ensuring robust performance and seamless integration with forecasting services.[www.ml-sol.com]
  - Effectively communicated complex machine learning concepts and model results to technical and non-technical stakeholders, improving cross-functional collaboration and decision-making.
  - Published research findings in Renewable Energy (Elsevier), contributing novel insights to solar PV forecasting literature.https://doi.org/10.1016/j.renene.2025.123021
  - Granted a patent based on the developed forecasting methodology, underscoring innovation and practical impact.

Skills Summary

Machine Learning for Energy Systems, Neural Networks and Deep Learning, Supervised Learning Algorithms, Time Series Forecasting, Predictive Modeling, Feature Engineering, Ensemble Learning, Model Optimization, Anomaly Detection, Data Preprocessing and Cleaning, Python for Data Analysis, Flask and Streamlit for Web Interface Development, Automation of Data Pipelines, Backend Development, Multi-objective Optimization, Solar PV System Modeling, Data-Driven Feasibility Analysis, Energy Yield Optimization, Power System Simulation (MATLAB/Simulink), Voltage Quality Improvement in Isolated Systems, Harmonic Distortion Mitigation, Electrical Substation Operations (132 kV), Switchgear & Protection, Industrial Automation (PLC, SCADA), Renewable Energy Systems Design, Numerical Analysis of Solar Thermal Systems.

Publications

- **R. Singh, R. Sharma, A. Rathore, and P. Gupta, "A system and method forecasting for solar photovoltaic power using machine-learning," Indian Patent 580682, granted Feb. 16, 2026.**:
- **A. Rathore, et al., "Day ahead solar forecast using long short term memory network augmented with Fast Fourier transform-assisted decomposition technique," *Renewable Energy 247 (2025): 123021.*:** https://doi.org/10.1016/j.renene.2025.123021
- **A. Rathore, et al., "Energy, exergy and performance analysis of a 380 kWP roof-top PV plant assisted with data-driven models for energy generation", *Journal of Thermal Engineering 10.5 (2024): 1164-1183*** :
- **A. Rathore et al., "An IoT-Based Automatic Light Control System for Smart Buildings", in *International Conference on Signal, Machines, Automation, and Algorithm*, Singapore: Springer Nature Singapore, 2023.**: https://doi.org/10.1007/978-981-97-6349-8$_4$2

- **Book Chapter: "Configurations of Converters applied to Electric Vehicles with comprehensive control strategies and optimization", accepted for publication in *Power Electronics for Electric Vehicles and Energy Storage: Emerging Technologies and Developments*, CRC Press, Taylor & Francis Group.**:
- **A. Rathore and S. Sundaram,"Building Energy Management Through Load Control Using Internet of Things" in *2021 Innovations in Power and Advanced Computing Technologies (i-PACT)*, 2021, pp. 1–6.**: doi:10.1109/i-PACT52855.2021.9696596

Projects and Training

- **M.Tech Thesis: "Yield Modelling and Data-Driven Feasibility Analysis of Solar PV and Solar Thermal Systems"**: Developed a data-driven model for solar PV energy yield forecasting with integrated optimization to maximize performance under varying climates. Enabled site-specific feasibility and parameter tuning for utility-scale and distributed installations using MATLAB and statistical techniques.
- **B.Tech Project: "Enhancement of Voltage Quality in Isolated Power Systems"**: Designed a battery-backed Series Compensator in MATLAB/Simulink to reduce Total Harmonic Distortion (THD) and improve voltage stability by actively compensating faults in isolated grids.
- **Industrial Training: Uttar Pradesh Power Corporation Limited (UPPCL), Lucknow**: Gained practical experience at a 132 kV substation, working with transformers, switchgear, lightning arresters, CT/PT devices, and grid protection protocols during live operations.
- **Workshop: Industrial Automation (Sofcon, SRMCEM, Lucknow)**: Hands-on training on PLC programming, HMI, SCADA, relay logic, and sensor integration for automation and process control.
- **Summer Training: UNICON PLC Automation**: Completed MATLAB-based training focused on control system fundamentals, dynamic system modeling, and signal processing applications relevant to power systems.

Courses and Certifications

- **Gen AI Foundational Models for NLP & Language Understanding**: (IBM/Coursera).
- **Generative AI and LLMs: Architecture and Data Preparation**: (IBM/Coursera).
- **Google Data Analytics certificate**: (Google/Coursera).
- **Programming for Everybody (Getting started with Python)** : (University of Michigan/Coursera).
- **Python Data Structures** : (University of Michigan/Coursera).
- **Intro to Machine Learning and Intermediate Machine Learning** : (Kaggle).
- **Neural Networks and Deep Learning** : (DeepLearning.AI/Coursera).
- **Self-paced training course on MATLAB and Simulink Onramp** : (MATLAB/MathWorks).
- **Self-paced training course on Machine learning** : (MATLAB/MathWorks).
- **Solving ordinary differential equations with MATLAB** : (MATLAB/Mathworks).
- **Machine Learning and Data Science Bootcamp** : (Udemy).

Honors and Awards

- **Institute Gold Medal**, Rajiv Gandhi Institute of Petroleum Technology (RGIPT) — Awarded for best academic performance in M.Tech Power and Energy Systems Engineering.
- **Certificate of Merit**, SRMCEM — Recognized for outstanding academic achievement during 2nd, 3rd, and 4th years of Bachelor of Technology.
- **Merit Certificate**, Central Board of Secondary Education (CBSE) — Awarded for exemplary performance in Class 12th examinations.

References

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